Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The second overtone of an open pipe has the same frequency as the first overtone of a closed pipe 2 m long. The length of the open pipe is

A. 8 m

B. 4 m

C. 2 m

D. 1 m

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $C_P - C_V$ is $1.00\ R$ for a gas sample in state A and is $1.08\ R$ in state B. Let $p_A,\ p_B$ denote the pressure and T_A and T_B denote the temperatures of states A and B, respectively. Most likely

A. $p_A < p_B$ and $T_A > T_B$

B. $p_A > p_B$ and $T_A < T_B$

C. $p_A = p_B$ and $T_A < T_B$

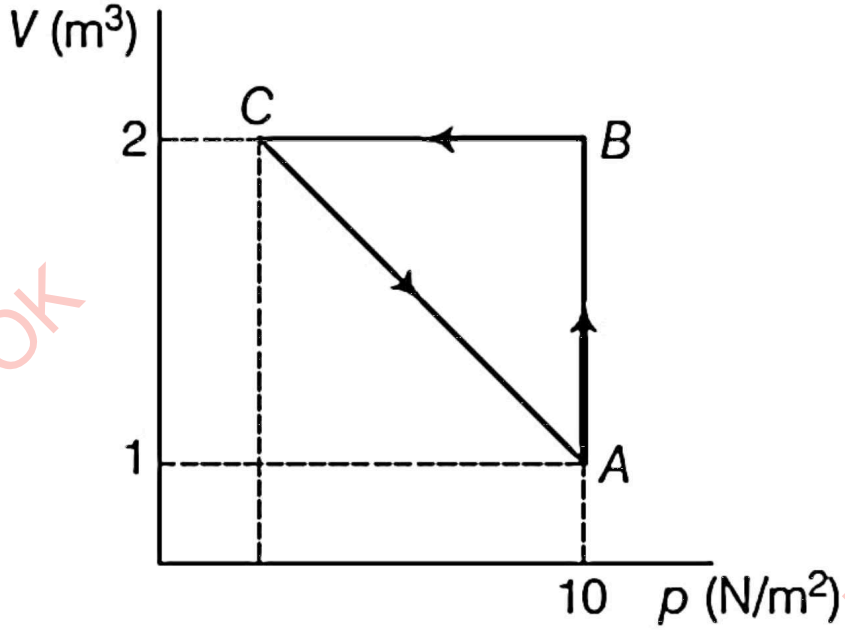
D. $p_A > p_B$ and $T_A = T_B$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An ideal gas is taken through a cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown. If the net heat supplied to the gas in the cycle is 5 J, then workdone by the gas in process $C \rightarrow A$ is



A. -5 J

B. -10 J

C. -15 J

D. -20 J

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The amplitude of two waves are in ratio $5:2$. If all other conditions for the two waves are same, then what is the ratio of their energy densities?

A. $5:2$

B. $5:4$

C. $4:5$

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D. $25 : 4$

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three closed vessels A, B and C are at the same temperature T and contain gases which obey the Maxwellian distribution of velocities. Vessel A contains only O_2, B only N_2 and C a mixture of equal quantities of O_2 and N_2. If the average speed of the O_2 molecules in vessel A is V_1, that of the N_2 molecules in vessel B is V_2, the average speed of the O_2 molecules in vessel C is

A. $(V_1 + V_2)/2$

B. V_1

C. $(V_1 V_2)^{1/2}$

D. $\sqrt{3kT/M}$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A rod of length l and cross-section area A has a variable thermal conductivity given by $K = \alpha T$, where α is a positive constant and T is the temperature in Kelvin. Two ends of the rod are maintained at temperatures T_1 and $T_2 (T_1 > T_2)$. Heat current flowing through the rod will be -

A. $\frac{A\alpha(T_1^2 - T_2^2)}{3l}$

B. $\frac{A\alpha(T_1^2 + T_2^2)}{l}$

C. $\frac{A\alpha(T_1^2 + T_2^2)}{3l}$

D. $\frac{A\alpha(T_1^2 - T_2^2)}{2l}$

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

10 mole of an ideal gas is heated at constant pressure of one atmosphere from $27^{\circ}\mathrm{C}$ to $127^{\circ}\mathrm{C}$. If $\mathrm{C_{v,m}} = 21.686 + 10^{-3}\ \mathrm{T}$, the $\Delta\mathrm{H}$ for the process is :

A. $3000\ \mathrm{J}$

B. $3350\ \mathrm{J}$

C. $3700\ \mathrm{J}$

D. $30350\ \mathrm{J}$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to the law $\frac{P}{V} = \text{constant}$. If the final temperature is $2T_0$, heat supplied to the gas is

A. $2RT_0$

B. RT_0

C. $\frac{3}{2}RT_0$

D. $\frac{1}{2}RT_0$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q, where V is the volume of the gas. The value of q is:
$\left(\gamma = \frac{C_P}{C_v}\right)$

A. $\frac{\gamma-1}{2}$

B. $\frac{3\gamma+1}{6}$

C. $\frac{3\gamma-5}{6}$

D. $\frac{\gamma+1}{2}$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The transverse displacement of a wave on string is given by $y(x,t) = e^{-(ax^2+bt^2+2\sqrt{ab}xt)}$, this represents a

A. Wave moving along $+x$-axis with speed $\sqrt{\frac{a}{b}}$

B. Wave moving along $-x$-axis with speed $\sqrt{\frac{b}{a}}$

C. Standing wave of frequency $\sqrt{b}$

D. Standing wave of frequency $\frac{1}{\sqrt{b}}$

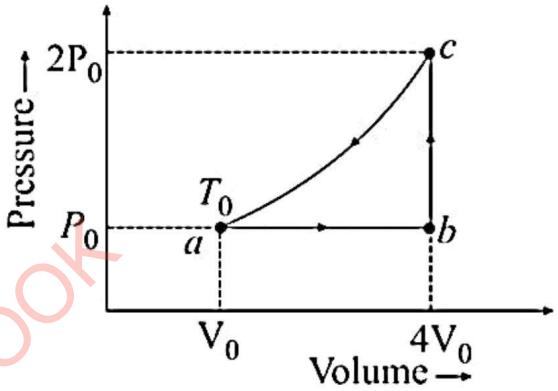
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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One mole of an ideal monoatomic gas is made to go through the cycle as shown in figure. Then the change in the internal energy in expanding the gas from a to c along the path abc is



A. $3P_0V_0$

B. $6RT_0$

C. $4.5RT_0$

D. $10.5RT_0$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A diatomic ideal gas is heated at constant volume until the pressure is doubled and again heated at constant pressure until the volume is doubled. The average molar heat capacity for the whole process is

A. $\frac{13R}{6}$

B. $\frac{19R}{6}$

C. $\frac{23R}{6}$

D. $\frac{17R}{6}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A system undergoes a reversible adiabatic process. The entropy of the system

A. Remains constant

B. May increase or may decrease

C. Increases

D. Decreases

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Statement - 1: Wave on the surface of a liquid is a combination of transverse and longitudinal motion of the liquid particles. But inside the liquid, the wave has to be longitudinal. and
Statement - 2: The bulk of liquid cannot have extensional tension or stress whereas the surface of liquid has property of extensional tension due to surface tension.

A. Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1

B. Statement - 1 is True, Statement - 2 is True; Statement - 2 is NOT a correct explanation for Statement - 1

C. Statement - 1 is True, Statement - 2 is False

Statement - 1 is True,
C. Statement - 2 is False

Statement - 1 is False,
D. Statement - 2 is True

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: According to kinetic theory of gases, all the collision between molecule-molecule, and between molecule-walls of container are perfectly elastic, so speed of each molecule in an isolated gaseous system remains constant.
Reason: Distribution of speed among the molecules of gas does not change with time for the above system.

If both Assertion and Reason are true and the Reason is correct explanation of the
A. Assertion.

If both Assertion and Reason are true but Reason is not correct explanation of the
B. Assertion.

If Assertion is true but
C. the Reason is false.

If Assertion is false but
D. Reason is true.

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Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements about the specific heat of a body is/are correct?
1. It depends upon the mass and shape of the body.
2. It is independent of the mass and shape of the body.
3. It depends only upon the temperature of the body.

Select the correct answer using the codes given below:

A. 2 and 3

B. 1 only

C. 1 and 3

D. 2 only

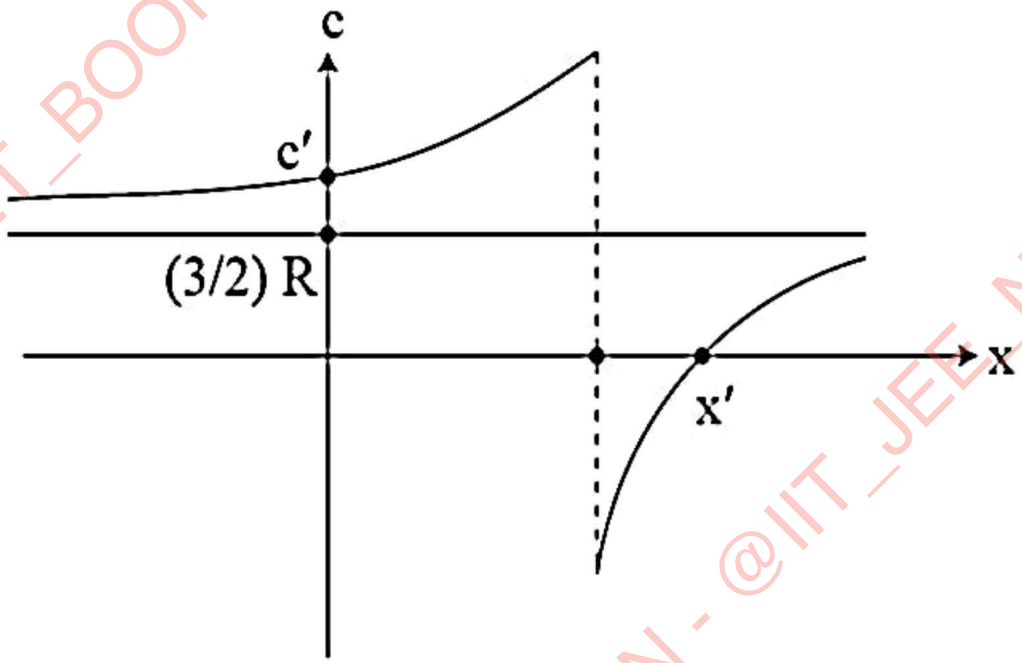
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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One mole of an ideal gas is taken along the process in which $PV^x = \text{constant}$. The graph shown represents the variation of molar heat capacity of such a gas with respect to x. The values of c' and x'. respectively, are given by



A. $\frac{5}{2}$ R, $\frac{5}{2}$

B. $\frac{5}{2}$ R, $\frac{5}{3}$

C. $\frac{7}{2}$ R, $\frac{7}{2}$

D. $\frac{5}{2}$ R, $\frac{7}{5}$

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two spherical black bodies have radii r_1 and r_2. Their surface temperatures are T_1 and T_2. If they radiate same power then $\frac{r_2}{r_1}$ is

A. $\frac{T_1}{T_2}$

B. $\frac{T_2}{T_1}$

C. $\left(\frac{T_1}{T_2}\right)^2$

D. $\left(\frac{T_2}{T_1}\right)^2$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A tuning fork is in resonance with a vibrating string under tension. When the tuning fork is moved with a certain velocity, 2 beats per second are heard. If the tension is increased, the beats disappear again. The direction of motion of the tuning fork is

A. towards the string

B. away from the string

C. random

D. unpredictable as data is not sufficient

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When 1 kg of ice at 0°C melts to water at 0°C the resulting change in its entropy, taking latent heat of ice to be 80 cal/g, calculate the change in entropy in this process ?

A. 273 cal/K

B. 8×10^{4} cal/K

C. 80 cal/K

D. 293 cal/K

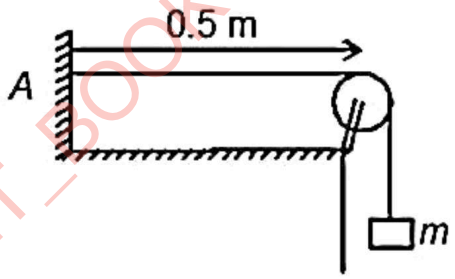
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A string having length 0.5 m fixed at one end and other end is connected to a block of mass $m = 2$ kg as shown in figure. The string is set into vibrations which is represented by $y = 4\sin\left(\frac{\pi x}{5}\right)\cos(50\pi t)$, where x and y are in cm and t is in second. The number of anti-nodes between point A and the fixed pulley is $2n$, find n.



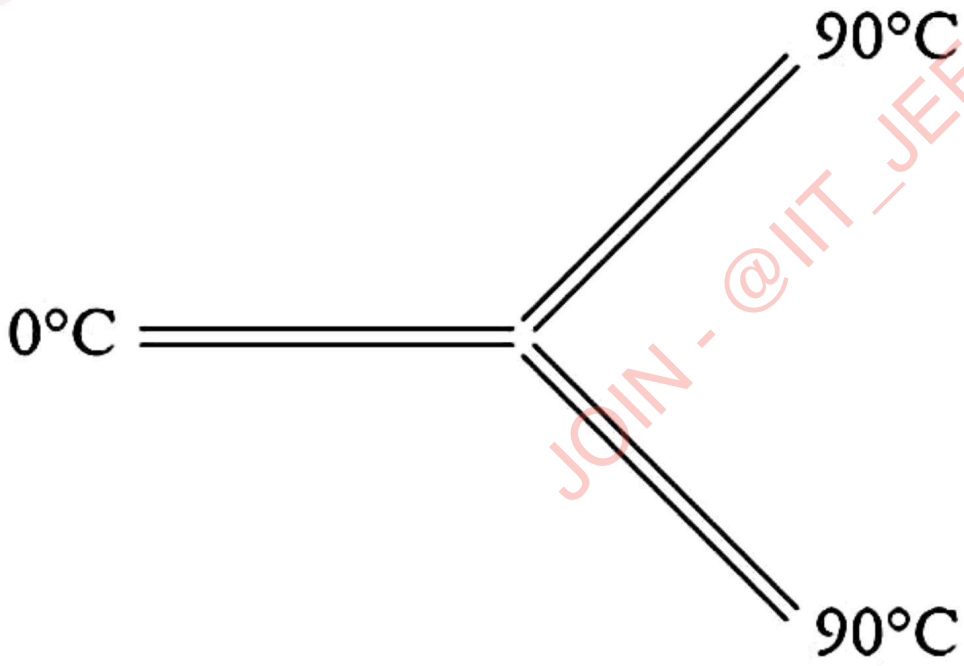
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Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Three rods made of the same material and having the same cross-section are joined as shown in the figure. Each rod is of the same length. What is the temperature at the junction of the three rods (in $^{\circ}C$)?



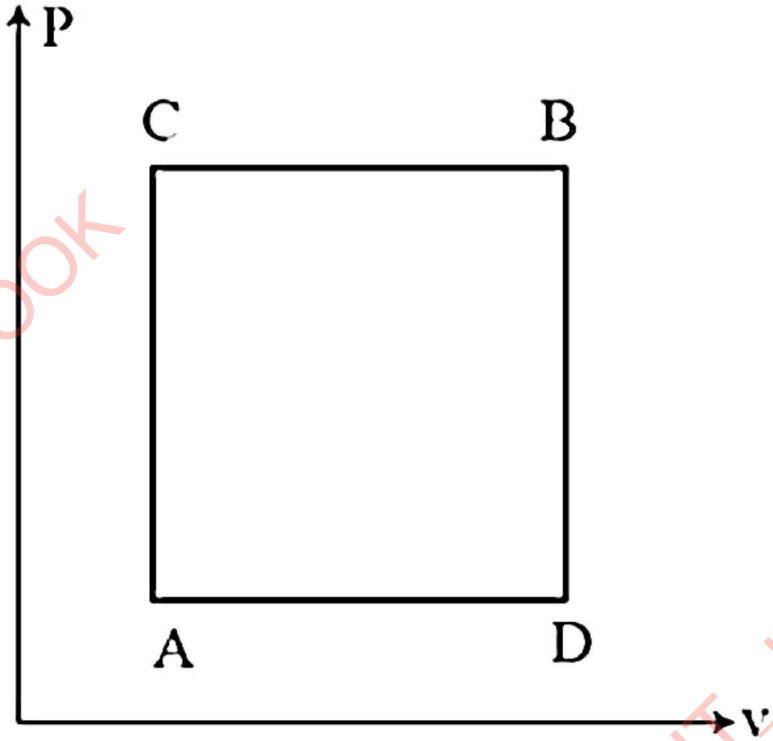
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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat flow (in joule) into the system in path ADB is :

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Consider a brass rod and a steel rod (80 cm longer than the brass rod) at $0^{\circ}C$. It is observed that on increasing temperatures of the two rods by same amount difference in lengths of the two rods does not change. Given that the thermal coefficient of linear expansion for steel and brass are $11 \times 10^{-6\circ}C^{-1}$ and $19 \times 10^{-6\circ}C^{-1}$ respectively. The sum of lengths of the two rods at $0^{\circ}C$ is (Mark answer in metre)

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Nitrogen gas is at $300\ ^{\circ}C$ temperature. The temperature (in K) at which the rms speed of a H_2 molecule would be equal to the rms speed of a nitrogen molecule, is ..... (Molar mass of N_2 gas 28 g and $1^{\circ}C = 273K$) (Round off to nearest integer).

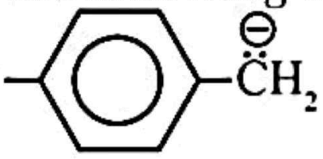
Chemistry Single Correct (Maximum Marks: 80)

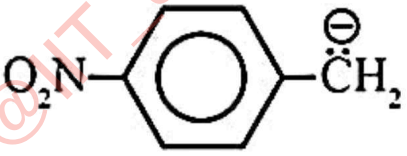
Question No. 1

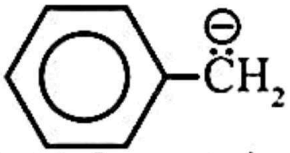
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the following carbanions:

(I) 

(II) 

(III) 

(IV) 

Correct decreasing order of stability is -

A. $II > III > IV > I$

B. $III > IV > I > II$

C. $IV > I > II > III$

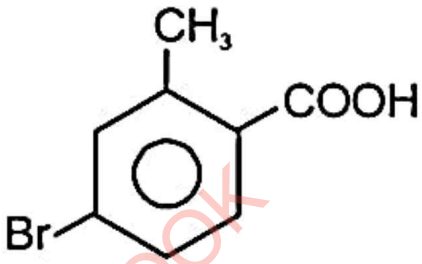
D. $I > II > III > IV$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A: o-Toluic acid on reaction with $Br_2 + Fe$ gives



R: The above reaction is followed by formation of carbanion.

A. If both Assertion & Reason are true and the reason is the correct explanation of the assertion, then mark (1).

B. If both Assertion & Reason are true but the reason is not the correct explanation of the assertion, then mark (2).

C. If Assertion is true statement but Reason is false, then mark (3).

D. If both Assertion and Reason are false statements, then mark (4).

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In Duma's method for estimation of nitrogen 0.30 g of organic compound gave 50 cm^3 of nitrogen gas collected at 300 K and 715 mm Hg pressure (Hint vapour pressure of water at 300 K is 15 mm Hg) then the value of percentage of nitrogen in the compound is

A. 17.46

B. 28.56

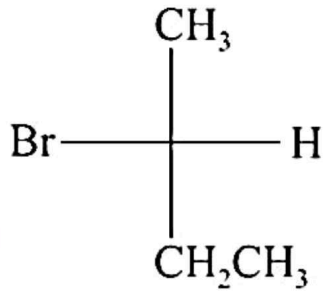
C. 28

D. 15.68

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When  reacts with KOH in the presence of water (through S_N2 reaction mechanism), then the stereochemistry of product so formed will be -

A. R

B. S

C. Mixture of R and S

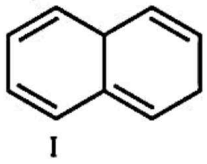
D. Partial S + racemic mixture

Question No. 5

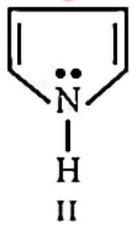
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

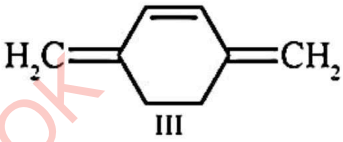
Which of the following will exhibit aromatic character?



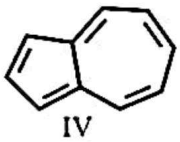
I



II



III



IV

A. I, II

B. III, IV

C. II, IV

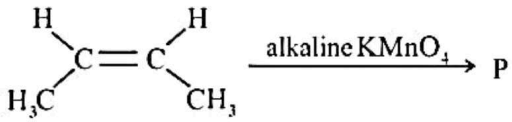
D. II, III

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is true for the reaction given below?



A. P is a the meso compound 2, 3-butanediol formed by syn-addition

B. P is a the meso compound 2, 3-butanediol formed by anti-addition

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P is a racemic mixture of (d) and (I) 2, 3-butanediol formed by

C. anti-addition

P is a racemic mixture of (d) and (I) 2, 3-butanediol formed by

D. syn-addition

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : Cis-1,3-dihydroxycyclohexane exists in boat conformation.

Reason : In the chair form, there will not be hydrogen bonding between the two hydroxyl groups.

If both assertion and reason are true and the reason is the correct explanation of the

A. assertion

If both assertion and reason are true and the reason is not the correct explanation of

B. the assertion

If assertion is true but

C. reason is false

If both assertion and

D. reason are false.

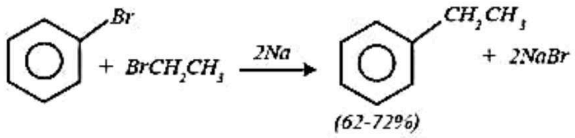
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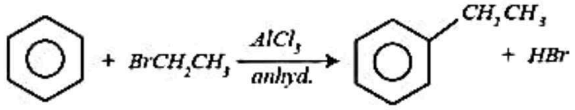
Question No. 8

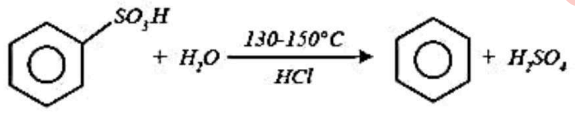
Only One Option Correct Type

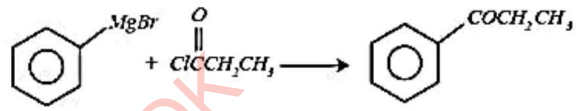
Each question has multiple options out of which ONLY ONE is correct.

Which of the following is Wurtz-Fittig reaction?

A. 

B. 

C. 

D. 

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The ammonia evolved from the treatment of 0.3 g of an organic compound to estimate nitrogen was passed into 100 mL 0.1 M sulphuric acid. The excess acid required 20 mL 0.5 M sodium hydroxide solution for complete neutralization. The organic compound is

A. acetamide.

B. benzamide.

C. urea.

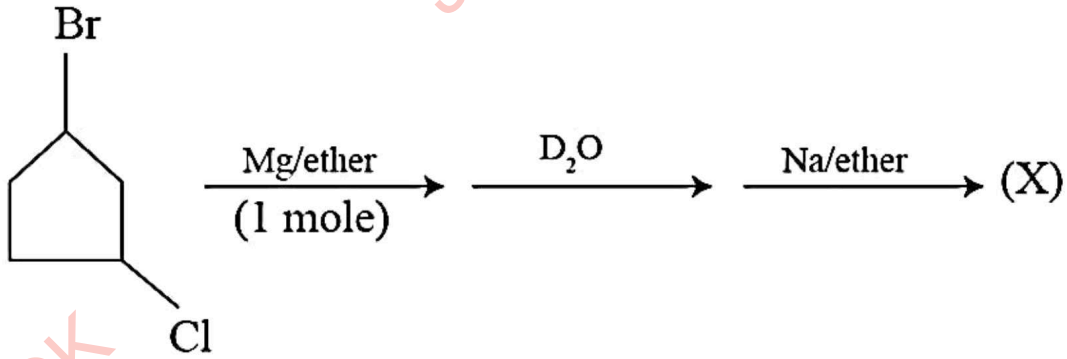
D. thio-urea.

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Question No. 10

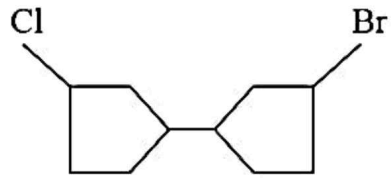
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

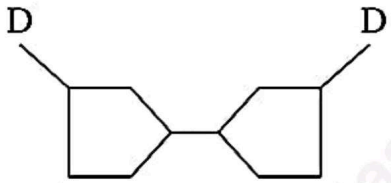


The compound (X) is

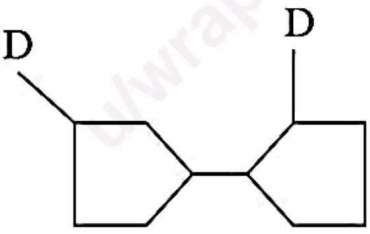
A.



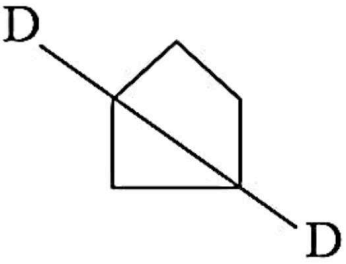
B.



C.



D.



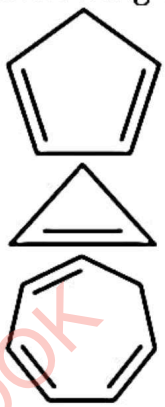
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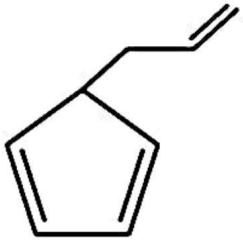
Question No. 11

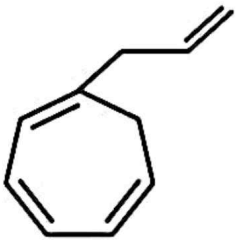
Only One Option Correct Type

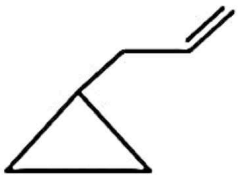
Each question has multiple options out of which ONLY ONE is correct.

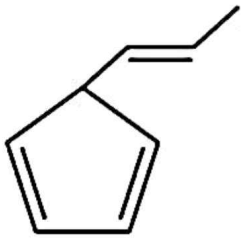
What will be the product when most acidic species among following will react with 3 −chloro prop−1 −ene?



A. 

B. 

C. 

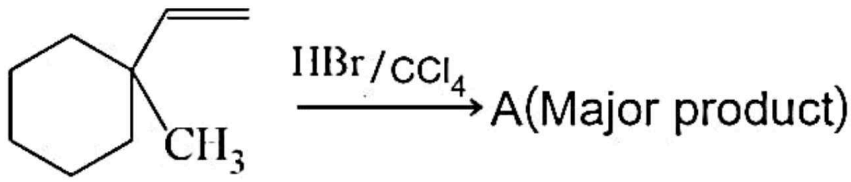
D. 

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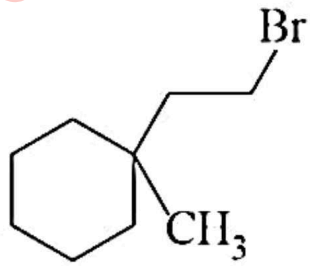
Question No. 12

Only One Option Correct Type

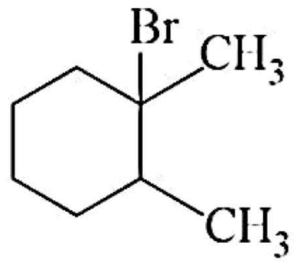
Each question has multiple options out of which ONLY ONE is correct.



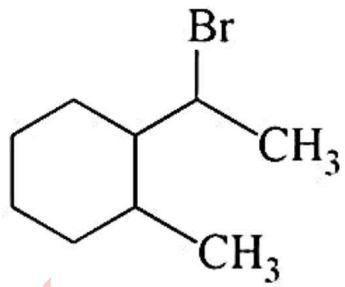
A.



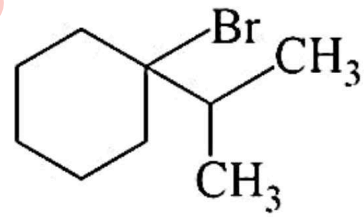
B.



C.



D.



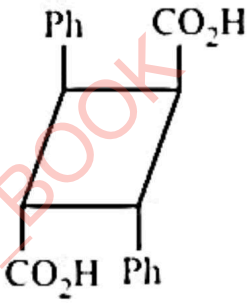
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Question No. 13

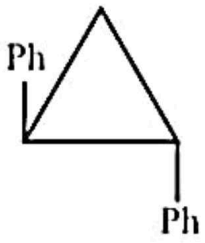
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

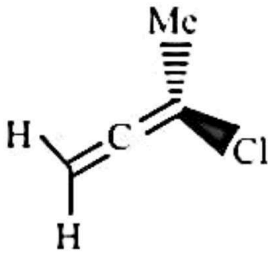
Among the following.



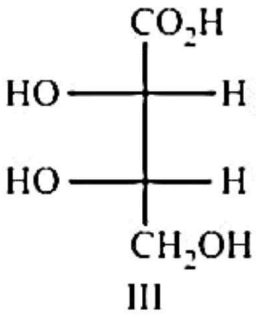
I



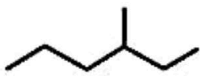
II



III



III



IV

the compounds which can exhibit optical activity are :

A. only II, IV and V

B. only IV and V

C. only I, II and V

D. only I, II and IV

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When bromocyclohexane and sodium propynide are heated together, the product will be
I. 2 −cyclohexylpropyne
II. 1 −cyclohexylpropyne
III. Propyne
IV. Cyclohexene

A. I and II

B. II, III and IV

C. I, II and II

D. All four products are formed

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements regarding the S_N1 reaction shown by alkyl halide is not correct?

A. the added nucleophile plays no kinetic role in S_N1 reaction

B. The S_N1 reaction involves the inversion of configuration of the optically active substrate

C. the S_N1 reaction on the chiral starting material end up with racemization of the product

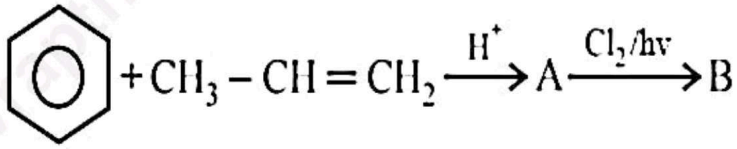
D. The more stable the carbocation intermediate the faster the S_N1 reaction

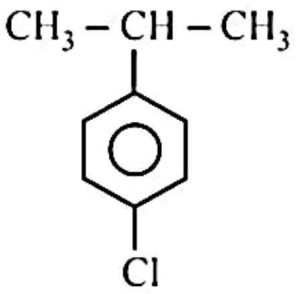
Question No. 16

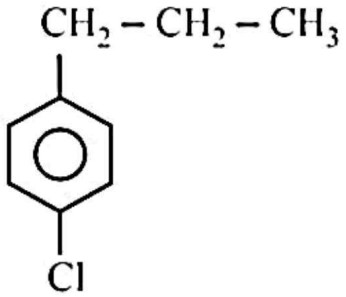
Only One Option Correct Type

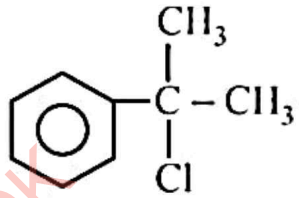
Each question has multiple options out of which ONLY ONE is correct.

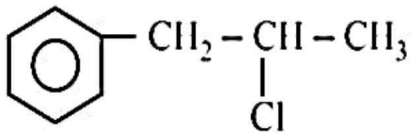
What is the product (major) B in the following reaction?



A. 

B. 

C. 

D. 

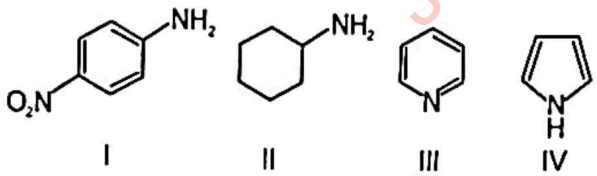
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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The order of basicity of



I II III IV

A. IV < III < I < II

B. II < I < IV < III

C. IV < I < III < II

D. II < III < I < IV

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Statement I : The 3-chloro-3, 4-dimethyl hex-1-ene undergo reaction with base to give butadiene.

Statement II : This is due to Sytzeff rule which states that more substituted product will be the more stable product.

A. Both Statement I and Statement II are true and the Statement II is the correct explanation of the Statement I.

B. Both Statement I and Statement II are true and the Statement II is not the correct explanation of the Statement I.

C. Statement I is false but Statement II is true.

D. Both Statement I and Statement II are false.

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following is a wrong statement about Oxymercuration-Demercuration?

A. In the 1st step oxymercuration occurs i.e., water and $Hg(OAc)_2$ add to double bond

B. In the 2nd step demercuration occurs i.e., $NaBH_4$ reduces $HgOAc$group to hydrogen

C. The net reaction is addition of water according to Markovnikov's rule

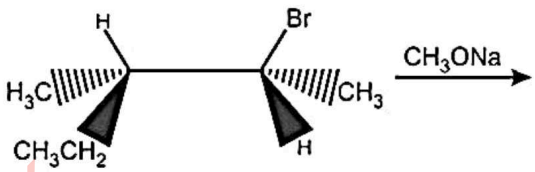
D. Rearrangement takes place

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Choose the correct statement about the major product formed in E2 reaction ?



A. The major product will be optically active

B. The major product will be trans-3-Methyl-2-pentene

C. The major product will be cis-3-Methyl-2-pentene

D. The major product will be 3-Methyl-1-pentene

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

How many isomers of $C_5H_{11}Br$ show stereochemical inversion when it undergoes SN^2 reaction?

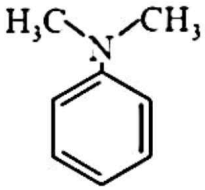
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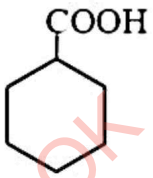
Question No. 2

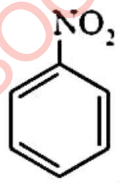
Numerical Type

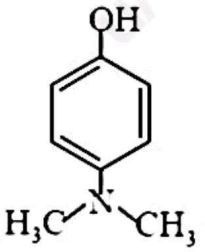
The answer has to be filled into the input box provided below.

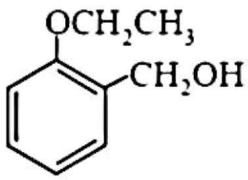
Amongst the following, the total number of compounds soluble in aqueous NaOH is

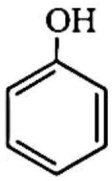


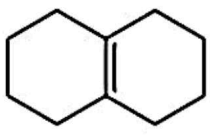


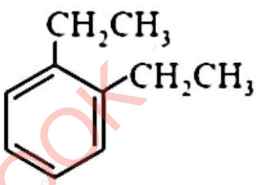


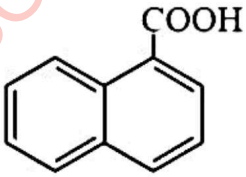












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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

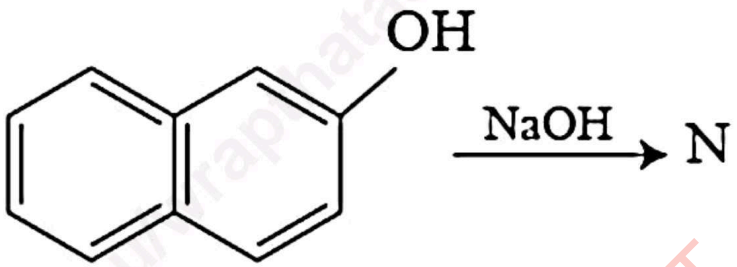
A hydrocarbon A, of the formula C_8H_{10}, on ozonolysis gives compound $B(C_4H_6O_2)$only. The Compound B can also be obtained from the alkyl bromide, $C(C_3H_5Br)$ upon treatment with magnesium in dry ether, followed by carbon dioxide and acidification. Give the number of secondary hydrogen atoms in compound A.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

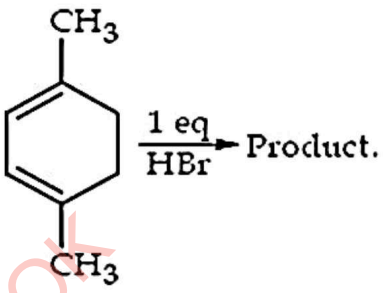
The number of resonance structures for N is



Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.



No of allylic hydrogens in thermodynamic product is "X"

No of allylic hydrogens atom in kinetic product is " Y "

The value of $|X + Y|$ is $\cdots$

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, where $i^2 = -1$. Let I denotes the identity matrix of order 2, then $I + A + A^2 + A^3 \ldots\ldots A^{110}$ is equal to

A. $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f : R \to R$ be a function such that $f(x) = x^3 + x^2 + 4x + \sin x$. Then, the function $f(x)$ is

A. one-one and onto

B. one-one and into

C. many-one and onto

D. many-one and into

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let A is a matrix of order 3×3 defined as $A = \left[a_{ij}\right]_{3\times 3}$, where $a_{ij} = \lim\limits_{x\to 0} \frac{1-\cos(ix)}{\sin(ix)\tan(jx)}$ $(\forall 1 \leq i, j \leq 3)$, then A^2 is equal to

A. A

B. $\frac{3}{2}A$

C. $\frac{2}{3}A$

D. $\frac{1}{4}A$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A,\ B$ and C are $n \times n$ matrices such that $|A| = 2,\ |B| = 3$ and $|C| = 5$. If $\left|(2A)^2(3B)(5C)^{-1}\right| = \frac{1728}{125}$, then the value of n is equal to

A. 2

B. 3

C. 4

D. 6

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f : R \to R$ is defined by

$$f(x) = \begin{cases} x+4 & \text{for} \quad x < -4 \\ 3x+2 & \text{for} \quad -4 \le x < 4 \\ x-4 & \text{for} \quad x \ge 4 \end{cases}$$

then the correct matching of List I from List-II is :

List-I	List-II
(A) $f(-5) + f(-4)$	(i) 14
(B) $f(\|f(-8)\|)$	(ii) 4
(C) $f(f(-7) + f(3))$	(iii) -11
(D) $f(f(f(f(0)))) + 1$	(iv) -1
	(v) 1
	(vi) 0

A. (A) (iii) (B) (vi) (C) (ii) (D) (v)

B. (A) (iii) (B) (iv) (C) (ii) (D) (v)

C. (A) (iv) (B) (iii) (C) (ii) (D) (i)

D. (A) (iii) (B) (vi) (C) (v) (D) (ii)

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The domain of the function $f(x) = \sqrt{\frac{4-x^2}{[x]+2}}$, where $[x]$ denotes the greatest integer not more than x, is

A. $(-\infty, -2) \cup (1, 2)$

B. $(-\infty, -2) \cup (-1, 2)$

C. $(-\infty, -2), \cup [-1, 2]$

D. $(-\infty, -1) \cup (1, 2)$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If A, B are two non-singular matrices of order 3 and I is an identity matrix of order 3 such that $AA^T = 5\mathrm{I}$ and $3A^{-1} = 2A^T - A\text{adj}(4B)$, then $|B|^2$ is equal to (where A^T and $\text{adj}(A)$ denote transpose and adjoint matrices of the matrix A respectively)

A. $\frac{7^3}{5^3 \cdot 4^6}$

B. $\frac{7^3}{5^3 \cdot 4^3}$

C. $\frac{7^4}{5^3 \cdot 2^{12}}$

D. $\frac{5^6}{7^3 \cdot 2^{10}}$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $S = \{(a, b, c) \in N \times N \times N : a + b + c = 21, a \leq b \leq c\}$ and $T = \{(a, b, c) \in N \times N \times N : a, b, c \text{ are in } AP\}$ where N is the set of all natural numbers. Then, the number of elements in the set $S \cap T$

A. 6

B. 7

C. 13

D. 14

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The range of the function $y = 2\sin^{-1}\left[x^2 + \frac{1}{2}\right] + \cos^{-1}\left[x^2 - \frac{1}{2}\right]$ is (where, $[\cdot]$ denotes the greatest integer function)

A. $(0, \pi)$

B. $\left[\pi, \frac{3\pi}{2}\right]$

C. $\{\pi\}$

D. $\left\{\pi, \frac{3\pi}{2}\right\}$

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the system of linear equations $x - 4y + 7z = g$; $3y - 5z = h$; $-2x + 5y - 9z = k$ is consistent, then:

A. $g + h + 2k = 0$

B. $g + 2h + k = 0$

C. $2g + h + k = 0$

D. $g + h + k = 0$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given two real sets $A = \{a, a_2, a_3 \cdots a_{2n}\}$ and $B\{b_1, b_2, \cdots b_n\}$. If $f : A \to B$ is a function such that every element of B has an inverse image and $f(a_1) \leq f(a_2) \leq f(a_3) \leq f(a_4) \cdots \leq f(a_{2n})$, then the number of such mappings are

A. ${}^{2n}C_n$

B. ${}^{2n}C_{n-1}$

C. ${}^{2n-1}C_{n-1}$

D. ${}^{2n+1}C_n$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(n-1)x & \cos(nx) & \cos(n+1)x \\ \sin(\mathrm{n}-1)\mathrm{x} & \sin(nx) & \sin(n+1)x \end{vmatrix}$ is zero, if

A. $\sin x = 0$

B. $\cos x = 0$

C. $a = 0$

D. $\cos x = \frac{1+a^2}{2a}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A box contains 9 slips bearing numbers $-3, -2, -1, 0, 1, 2, 3, 4$ and 5. An experiment consists of drawing a slip from this box and replacing it back in the box after noting the number. This experiment is repeated 9 times. These 9 numbers are now chosen as elements of a 3×3 matrix, then the probability that the matrix is skew symmetric is

A. $\frac{1}{9^6}$

B. $\frac{343}{9^9}$

C. $\frac{1}{9^9}$

D. $\frac{1}{9^7}$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let P be the relation defined on the set of all real numbers such that $\mathrm{P} = \{(\mathrm{a}, \mathrm{b}) : \sec^2\mathrm{a} - \tan^2\mathrm{b} = 1\}$. Then P is

A. reflexive and symmetric but not transitive

B. symmetric and transitive but not reflexive

C. reflexive and transitive but not symmetric

D. an equivalence relation

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let A be a square matrix of order two with $|A| \neq 0$, such that $|A + |A|\mathrm{adj}(A)| = 0$, where $\mathrm{adj}(A)$ is an adjoint of matrix A, then the value of $|A - |A|\mathrm{adj}(A)|$ is

A. 1

B. 2

C. 3

D. 4

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f : R - \{3\} \to R - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g : R \to R$ be given as $g(x) = 2x - 3$. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to

A. 7

B. 2

C. 5

D. 3

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A and B are symmetric matrices of the same order. If $X = AB + BA$, $Y = AB - BA$ and $(XY)^T$ is equal to

A. XY

B. YX

C. $-YX$

D. None of these

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The existence of the unique solution of the system of equations $2x + y + z = \beta, 10x - y + \alpha z = 10$ and $4x + 3y - z = 6$ depends on

A. Both α and β

B. Neither β nor α

C. β only

D. α only

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = (a_{ij})_{3\times3}$ and $B = (b_{ij})_{3\times3}$, where $b_{ij} = \frac{a_{ij}+a_{ji}}{2}\ \forall i, j$. Number of such matrices A whose elements are selected from the set $\{0, 1, 2, 3\}$ such that $A = B$, are

A. 2^9

B. 2^{12}

C. 2^6

D. 2^8

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f : R \to R$ be defined by $f(x) = x^2 - \frac{x^2}{1+x^2}$ for all $x \in R$. Then,

A. f is one-one but not onto mapping

B. f is onto but not one-one mapping

C. f is both one-one and onto

D. f is neither one-one nor onto

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The number of all 3×3 matrices A, with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3, is _____.

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Let $f(x) = \frac{9^x}{9^x+3}$ and $f(x) + f(1-x) = 1$ then find value of $f\left(\frac{1}{1996}\right) + \left(\frac{2}{1996}\right) + \ldots\ldots + f\left(\frac{1995}{1996}\right)$. (mark your answer multiply by 10)

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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Let M be a square matrix of order 3 whose elements are real numbers and $adj(adjM) = \begin{bmatrix} 36 & 0 & -4 \\ 0 & 6 & 0 \\ 0 & 3 & 6 \end{bmatrix}$, then the absolute value of $Tr(M)$ is

[Here, $adjP$ denotes adjoint matrix of P and $T_r(P)$ denotes trace of matrix P i.e. sum of all principal diagonal elements of matrix P]

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A survey of 500 television viewers produced the following information, 285 watch football, 195 watch hockey, 115 watch basket-ball, 45 watch football and basket ball, 70 watch football and hockey, 50 watch hockey and basket ball, 50 do not watch any of the three games. The number of viewers, who watch exactly one of the three games are

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Let $[x]$ be the greatest integer less than or equal to x, for a real number x. Then the following sum $\left[\frac{2^{2020}+1}{2^{2018}+1}\right] + \left[\frac{3^{2020}+1}{3^{2018}+1}\right] + \left[\frac{4^{2020}+1}{4^{2018}+1}\right] + \left[\frac{5^{2020}+1}{5^{2018}+1}\right] + \left[\frac{6^{2020}+1}{6^{2018}+1}\right]$ is :-